Sourcing File

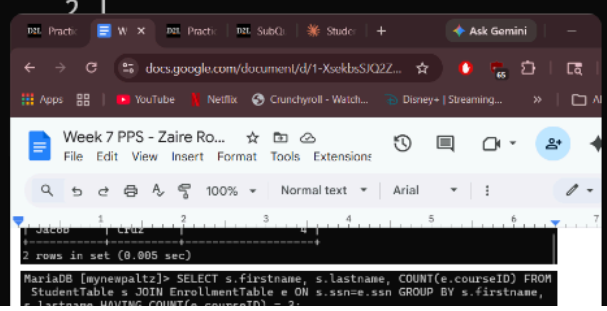
```
MySQL Client (MariaDB 12.1 (>

ERROR: Failed to open file '"C:\\Users\PHOENIX\OneDrive\Documents\SCHOOL\pri
nciples\Populate_Database_mynewpaltz.sql"', error: 13
MariaDB [(none)]> SOURCE C:\\Users\\PHOENIX\\OneDrive\\Documents\\SCHOOL\\pr
inciples\\Populate_Database_mynewpaltz.sql
Query OK, 0 rows affected, 1 warning (0.012 sec)

Database changed
Query OK, 0 rows affected, 1 warning (0.009 sec)

Query OK, 0 rows affected, 1 warning (0.002 sec)

Query OK, 0 rows affected, 1 warning (0.001 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 19 rows affected (0.065 sec)

Query OK, 11 rows affected (0.003 sec)

Query OK, 8 rows affected (0.001 sec)

Query OK, 0 rows affected (0.000 sec)

Query OK, 100 rows affected (0.005 sec)
Records: 100  Duplicates: 0  Warnings: 0

Query OK, 5 rows affected (0.001 sec)
Records: 5  Duplicates: 0  Warnings: 0

Query OK, 5 rows affected (0.002 sec)
Records: 5  Duplicates: 0  Warnings: 0

Query OK, 5 rows affected (0.001 sec)
Records: 5  Duplicates: 0  Warnings: 0

Query OK, 5 rows affected (0.001 sec)
Records: 5  Duplicates: 0  Warnings: 0

Query OK, 251 rows affected (0.002 sec)
Records: 251  Duplicates: 0  Warnings: 0
```

1. List students by name and ssn who are taking 5 courses
2. List students by name and ssn who are taking 4 courses
3. List students by name and ssn who are taking 3 courses
4. List students by name and ssn who are taking 2 courses
5. List students by name and ssn who are taking 1 course
6. List students by name and ssn who are taking 0 courses

```
MariaDB [mynewpaltz]> SELECT s.firstname, s.lastname, COUNT(e.courseID) FROM
 StudentTable s JOIN EnrollmentTable e ON s.ssn=e.ssn GROUP BY s.firstname,
s.lastname HAVING COUNT(e.courseID) = 4;
+-----------+----------+------------------+
| firstname | lastname | COUNT(e.courseID) |
+-----------+----------+------------------+
| Amanda    | Rivera   |                4 |
| Jacob     | Cruz     |                4 |
+-----------+----------+------------------+
2 rows in set (0.005 sec)
```

```
MariaDB [mynewpaltz]> SELECT s.firstname, s.lastname, COUNT(e.courseID) FROM
 StudentTable s JOIN EnrollmentTable e ON s.ssn=e.ssn GROUP BY s.firstname,
s.lastname HAVING COUNT(e.courseID) = 3;
+-------------+-----------+------------------+
| firstname   | lastname  | COUNT(e.courseID) |
+-------------+-----------+------------------+
| Barbara     | Lopez     |                3 |
| Christopher | Jackson   |                3 |
| David       | Rodriguez |                3 |
| Elizabeth   | Martinez  |                3 |
| James       | Smith     |                3 |
| Jennifer    | Garcia    |                3 |
| Jessica     | Thomas    |                3 |
| John        | Jones     |                3 |
| Joseph      | Anderson  |                3 |
| Karen       | Martin    |                3 |
| Linda       | Davis     |                3 |
| Mary        | Johnson   |                3 |
| Michael     | Miller    |                3 |
| Patricia    | Brown     |                3 |
| Richard     | Gonzalez  |                3 |
| Robert      | Williams  |                3 |
| Sarah       | Moore     |                3 |
| Susan       | Wilson    |                3 |
| Thomas      | Taylor    |                3 |
| William     | Hernandez |                3 |
+-------------+-----------+------------------+
20 rows in set (0.002 sec)
```

```
MariaDB [mynewpaltz]> SELECT s.firstname, s.lastname, COUNT(e.courseID) FROM
 StudentTable s JOIN EnrollmentTable e ON s.ssn=e.ssn GROUP BY s.firstname,
s.lastname HAVING COUNT(e.courseID) = 2;
+-----------+------------+-------------------+
| firstname | lastname   | COUNT(e.courseID) |
+-----------+------------+-------------------+
| Aaron     | West       |                 2 |
| Alexander | Diaz       |                 2 |
| Amy       | Murphy     |                 2 |
| Angela    | Ortiz      |                 2 |
| Anna      | Bailey     |                 2 |
| Benjamin  | Alvarado   |                 2 |
| Brandon   | Howard     |                 2 |
| Brenda    | Kelly      |                 2 |
| Catherine | Graham     |                 2 |
| Christine | Alexander  |                 2 |
| Debra     | Griffin    |                 2 |
| Dennis    | Hamilton   |                 2 |
| Diane     | Jordan     |                 2 |
| Emma      | Washington |                 2 |
| Eric      | Morales    |                 2 |
| Frank     | Kim        |                 2 |
| Gregory   | Butler     |
| Heather   | Cole       |
| Helen     | Cooper     |
| Janet     | Ford       |
| Jerry     | Sullivan   |
| Jonathan  | Gutierrez  |
| Justin    | Peterson   |
| Katherine | Gonzales   |
| Kathleen  | Morris     |
```



```
MariaDB [mynewpaltz]> SELECT s.firstname, s.lastname, COUNT(e.courseID) FROM
 StudentTable s JOIN EnrollmentTable e ON s.ssn=e.ssn GROUP BY s.firstname,
s.lastname HAVING COUNT(e.courseID) = 1;
+-----------+----------+-------------------+
| firstname | lastname | COUNT(e.courseID) |
+-----------+----------+-------------------+
| Andrew    | Wright   |                 1 |
| Anthony   | Harris   |                 1 |
| Ashley    | Young    |                 1 |
| Betty     | Sanchez  |                 1 |
| Charles   | Lee      |                 1 |
| Donald    | Lewis    |                 1 |
| Donna     | Flores   |                 1 |
| Dorothy   | King     |                 1 |
| Emily     | Nguyen   |                 1 |
| Joshua    | Torres   |                 1 |
| Kenneth   | Hill     |                 1 |
| Kimberly  | Scott    |                 1 |
| Lisa      | White    |                 1 |
| Margaret  | Ramirez  |                 1 |
| Mark      | Clark    |                 1 |
| Matthew   | Thompson |                 1 |
| Nancy     | Perez    |                 1 |
| Paul      | Allen    |                 1 |
| Sandra    | Robinson |                 1 |
| Steven    | Walker   |                 1 |
+-----------+----------+-------------------+
20 rows in set (0.002 sec)
```

```
MariaDB [mynewpaltz]> SELECT s.firstname, s.lastname, COUNT(e.courseID) FROM
 StudentTable s JOIN EnrollmentTable e ON s.ssn=e.ssn GROUP BY s.firstname,
s.lastname HAVING COUNT(e.courseID) = 0;
Empty set (0.002 sec)
```

7. Give a count of 4 credit courses by subjectID (# of 4-credit courses in each subject).
8. Give a count of 3 credit courses by subjectID
9. Give a count of 2 credit courses by subjectID

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(courseID) FROM CourseTable WHE
RE numOfCredits=4 GROUP BY subjectID;
+-----------+-----------------+
| subjectID | COUNT(courseID) |
+-----------+-----------------+
| CS        |               2 |
| ENGR      |               2 |
| MATH      |               2 |
+-----------+-----------------+
3 rows in set (0.002 sec)

MariaDB [mynewpaltz]>
```

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(courseID) FROM CourseTable WHE
RE numOfCredits=3 GROUP BY subjectID;
+-----------+-----------------+
| subjectID | COUNT(courseID) |
+-----------+-----------------+
| CS        |               3 |
| ENGL      |               5 |
| ENGR      |               2 |
| MATH      |               3 |
+-----------+-----------------+
4 rows in set (0.001 sec)

MariaDB [mynewpaltz]>
```

```
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(courseID) FROM CourseTable WHE
RE numOfCredits=2 GROUP BY subjectID;
+-----------+-----------------+
| subjectID | COUNT(courseID) |
+-----------+-----------------+
| ENGR      |               1 |
+-----------+-----------------+
1 row in set (0.001 sec)

MariaDB [mynewpaltz]>
```

10. Are there any 5 credit courses?Are there any 1 credit courses?
NO

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MariaDB server version for the right syntax to use n
ear ')' at line 1
MariaDB [mynewpaltz]> SELECT subjectID, COUNT(courseID) FROM CourseTable WHE
RE numOfCredits=5 GROUP BY subjectID;
Empty set (0.001 sec)

MariaDB [mynewpaltz]> SELECT subjectID, COUNT(courseID) FROM CourseTable WHE
RE numOfCredits=1 GROUP BY subjectID;
Empty set (0.001 sec)

MariaDB [mynewpaltz]>
```

11. List all enrolled students by name and their grades in each enrolled course

```
MariaDB [mynewpaltz]> SELECT firstName, lastName, grade, title FROM StudentT
able s LEFT JOIN EnrollmentTable e ON s.ssn=e.ssn JOIN CourseTable c ON e.co
urseID=c.courseID;
+-------------+------------+-------+----------------------------+
| firstName   | lastName   | grade | title                      |
+-------------+------------+-------+----------------------------+
| James       | Smith      | A     | Calculus I                 |
| Jennifer    | Garcia     | B     | Calculus I                 |
| William     | Hernandez  | B     | Calculus I                 |
| Jessica     | Thomas     | C     | Calculus I                 |
| Charles     | Lee        | A     | Calculus I                 |
| Kevin       | Green      | A     | Calculus I                 |
| Edward      | Hall       | B     | Calculus I                 |
| Timothy     | Carter     | A     | Calculus I                 |
| Cynthia     | Reyes      | A     | Calculus I                 |
| Anna        | Bailey     | B     | Calculus I                 |
| Frank       | Kim        | C     | Calculus I                 |
| Janet       | Ford       | A     | Calculus I                 |
| Tyler       | Woods      | A     | Calculus I                 |
| Mary        | Johnson    | B     | Calculus II                |
| Michael     | Miller     | A     | Calculus II                |
| Barbara     | Lopez      | A     | Calculus II                |
| Thomas      | Taylor     | B     | Calculus II                |
| Nancy       | Perez      | B     | Calculus II                |
| Kevin       | Green      | B     | Calculus II                |
| Amanda      | Rivera     | A     | Calculus II                |
| Deborah     | Roberts    | B     | Calculus II                |
| Sharon      | Edwards    | A     | Calculus II                |
| Scott       | Reed       | C     | Calculus II                |
| Nicole      | Foster     | B     | Calculus II                |
| Dennis      | Hamilton   | B     | Calculus II                |
| Heather     | Cole       | C     | Calculus II                |
| Robert      | Williams   | A     | Linear Algebra             |
| Linda       | Davis      | C     | Linear Algebra             |
| Richard     | Gonzalez   | B     | Linear Algebra             |
| Sarah       | Moore      | A     | Linear Algebra             |
| Matthew     | Thompson   | C     | Linear Algebra             |
| Kevin       | Green      | A     | Linear Algebra             |
```

12. For the course identified by courseID of 10, list the number of students enrolled, course number as well as subjectID.

```
MariaDB [mynewpaltz]> SELECT COUNT(e.ssn) AS Students, courseNumber, subject
ID FROM CourseTable c JOIN EnrollmentTable e ON c.courseID=e.courseID WHERE
e.courseID=10;
+----------+--------------+-----------+
| Students | courseNumber | subjectID |
+----------+--------------+-----------+
|       13 |          450 | CS        |
+----------+--------------+-----------+
1 row in set (0.004 sec)

MariaDB [mynewpaltz]>
```

13. How many students are in each course (identified uniquely by courseID)?

```
MariaDB [mynewpaltz]> SELECT COUNT(s.ssn) AS students, title, c.courseID FRO
M StudentTable s JOIN EnrollmentTable e ON s.ssn=e.ssn JOIN CourseTable c ON
 e.courseID=c.courseID GROUP BY c.courseID;
+----------+----------------------------+----------+
| students | title                      | courseID |
+----------+----------------------------+----------+
|       13 | Calculus I                 |        1 |
|       13 | Calculus II                |        2 |
|       12 | Linear Algebra             |        3 |
|       14 | Probability and Statistics |        4 |
|       13 | Abstract Algebra           |        5 |
|       11 | Intro to Computer Science  |        6 |
|       13 | Data Structures            |        7 |
|       13 | Operating Systems          |        8 |
|       12 | Database Management Systems |        9 |
|       13 | Artificial Intelligence    |       10 |
|       14 | English Composition I      |       11 |
|       13 | English Composition II     |       12 |
|       14 | World Literature           |       13 |
|       14 | Creative Writing           |       14 |
|       12 | Shakespeare                |       15 |
|       12 | Intro to Engineering       |       16 |
|       12 | Statics                    |       17 |
|       12 | Dynamics                   |       18 |
|       11 | Thermodynamics             |       19 |
|       10 | Circuit Analysis           |       20 |
+----------+----------------------------+----------+
20 rows in set (0.001 sec)

MariaDB [mynewpaltz]>
```

14. For each student enrolled, list the first name, last name, and number of credits that student was taking.

```
MariaDB [mynewpaltz]> SELECT firstName, lastName, SUM(numOfCredits) AS credi
ts FROM StudentTable s LEFT JOIN EnrollmentTable e ON s.ssn=e.ssn JOIN Cours
eTable c ON e.courseID=c.courseID GROUP BY s.firstName;
+-------------+------------+---------+
| firstName   | lastName   | credits |
+-------------+------------+---------+
| Aaron       | West       |       7 |
| Alexander   | Diaz       |       6 |
| Amanda      | Rivera     |      15 |
| Amy         | Murphy     |       7 |
| Andrew      | Wright     |       3 |
| Angela      | Ortiz      |       6 |
| Anna        | Bailey     |       6 |
| Anthony     | Harris     |       3 |
| Ashley      | Young      |       3 |
| Barbara     | Lopez      |      10 |
| Benjamin    | Alvarado   |       7 |
| Betty       | Sanchez    |       4 |
| Brandon     | Howard     |       7 |
| Brenda      | Kelly      |       6 |
| Brian       | Nelson     |      15 |
| Carol       | Baker      |      16 |
| Catherine   | Graham     |       6 |
| Charles     | Lee        |       4 |
| Christine   | Alexander  |       6 |
| Christopher | Jackson    |       9 |
| Cynthia     | Reyes      |      16 |
| David       | Rodriguez  |      10 |
| Deborah     | Roberts    |      17 |
| Debra       | Griffin    |       6 |
| Dennis      | Hamilton   |       7 |
| Diane       | Jordan     |       6 |
| Donald      | Lewis      |       3 |
| Donna       | Flores     |       4 |
| Dorothy     | King       |       3 |
| Edward      | Hall       |      16 |
| Elizabeth   | Martinez   |      10 |
| Emily       | Nguyen     |       3 |
| Emma        | Washington |       6 |
| Eric        | Morales    |       6 |
```